

Lemma XII states that all parallelograms circumscribed about conjugate diameters of an ellipse are equal, giving the constant-area bridge $CD^2 : CB^2$. Through this the semi-diameters that fill the messy ratio chain collapse onto the fixed axis constants. The decisive quantity is the latus rectum $L = 2BC^2/AC$, the chord through the focus S perpendicular to the major axis. Newton assembles $L \cdot QR : QT^2$, and Lemma XII lets every ellipse-shape factor cancel against another, leaving only SP^2 .